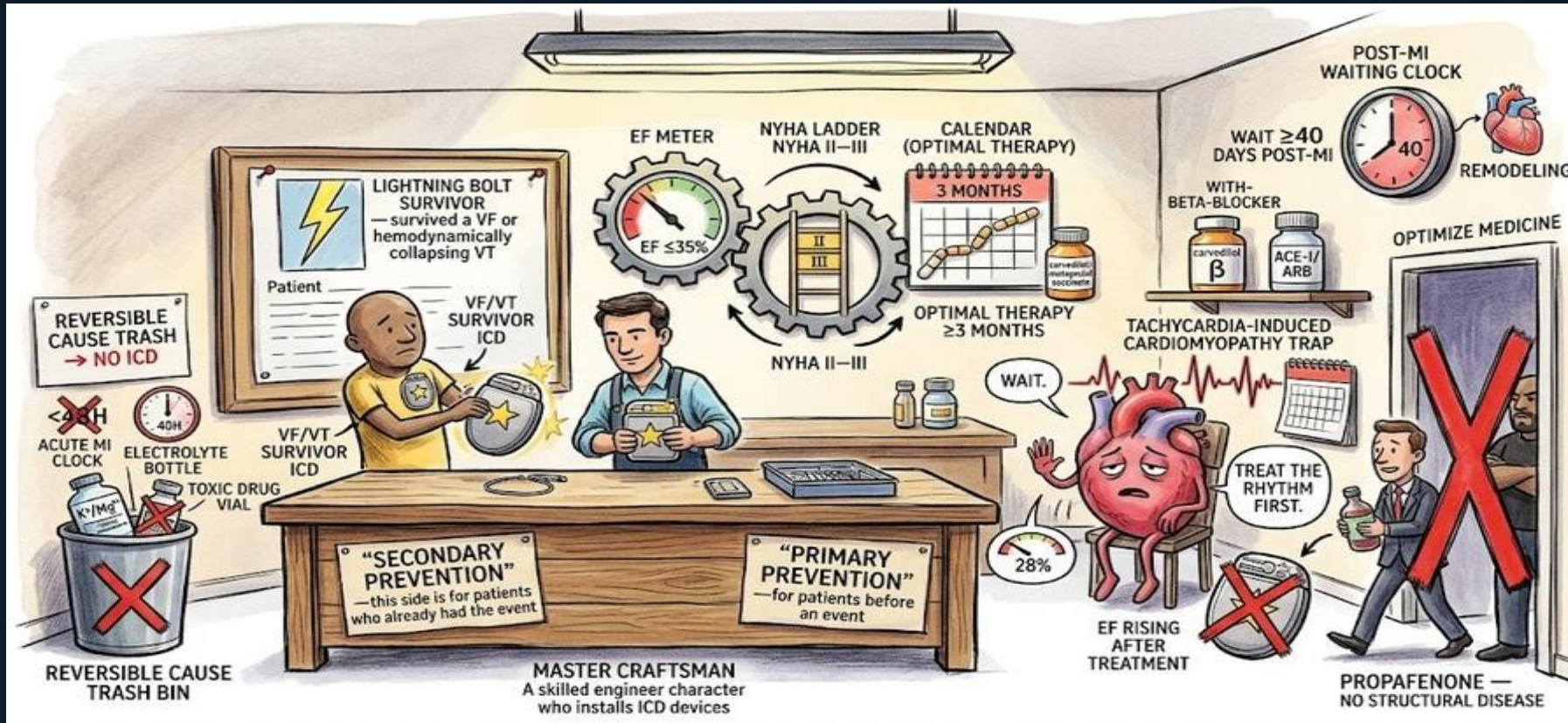
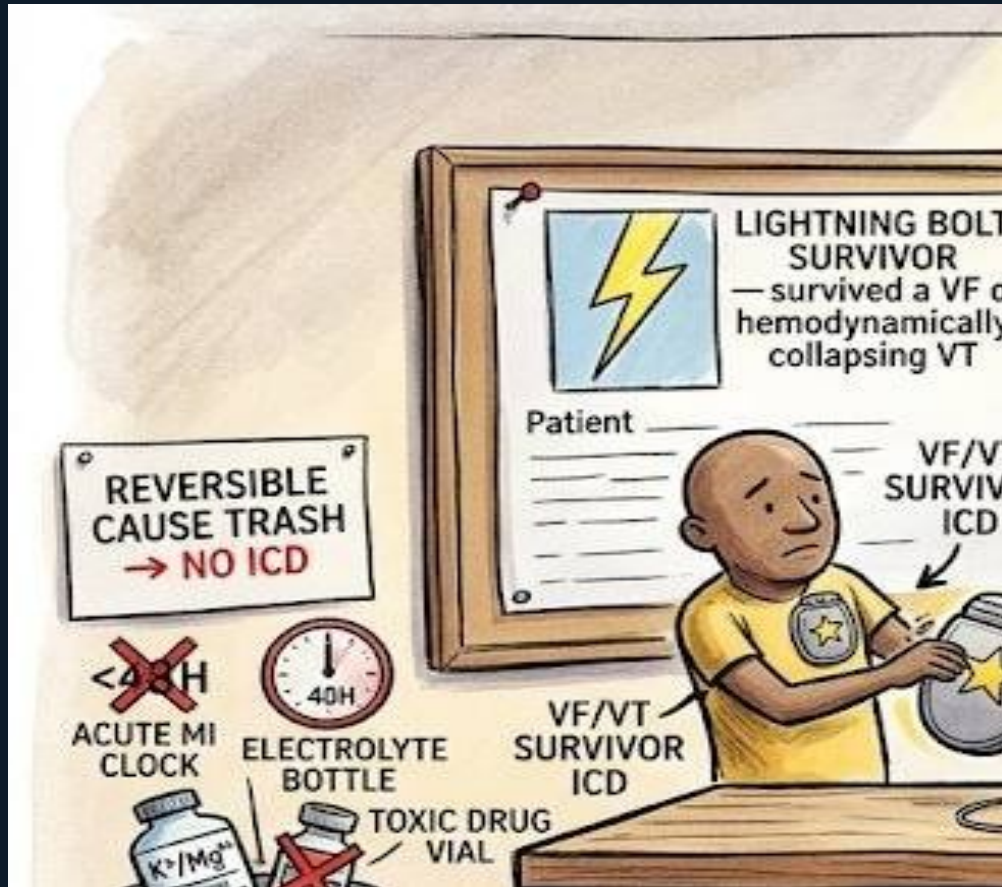


Scene E: VT & ICD Indications





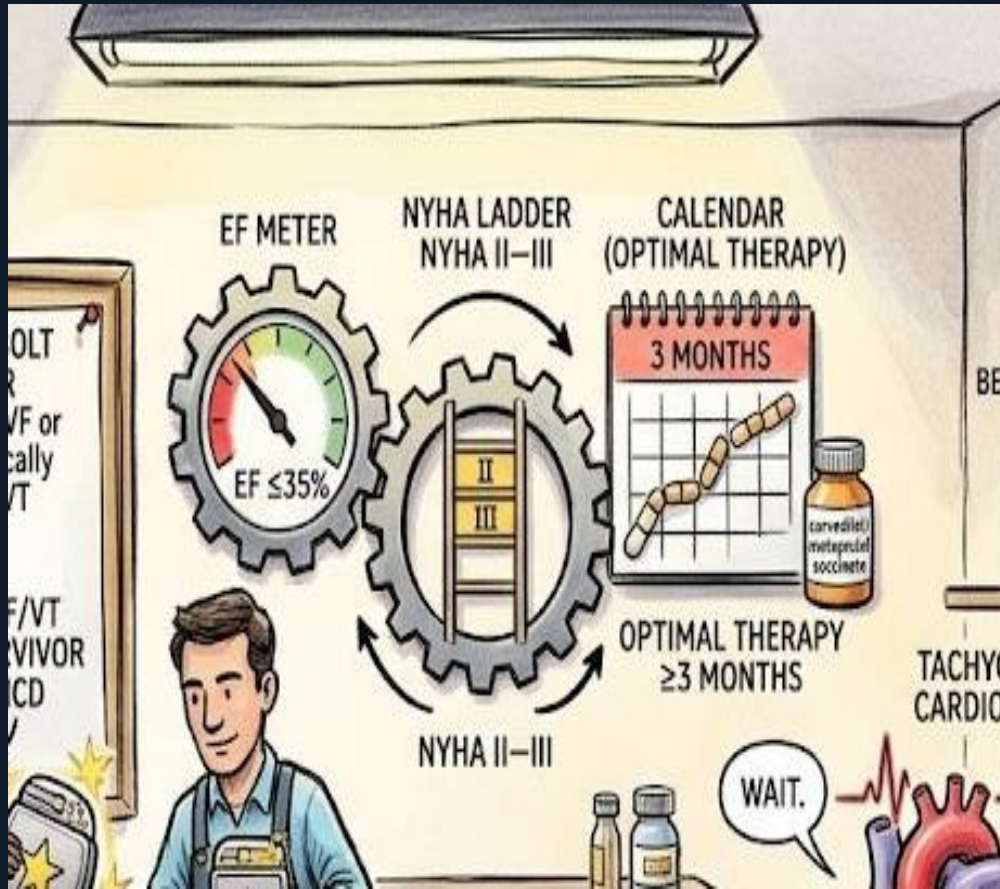
Scene E — VT & ICD Indications | Arrhythmias Board Prep

Secondary Prevention — Survived the Event

A lightning bolt survivor poster shows a patient who survived VF or hemodynamically collapsing VT. The craftsman installs an ICD immediately with a gold star. A reversible cause trash bin shows items with red Xs: acute MI clock (<48h), electrolyte bottle, toxic drug vial.

Secondary prevention ICD: any patient who survived VF or hemodynamically significant VT not due to a reversible or transient cause → ICD regardless of EF.

Board trap: patient with VT in the setting of acute hypokalemia → correct the electrolyte first. If VT recurs or no reversible cause found → ICD. The trash bin items are the reversible causes that must be excluded.

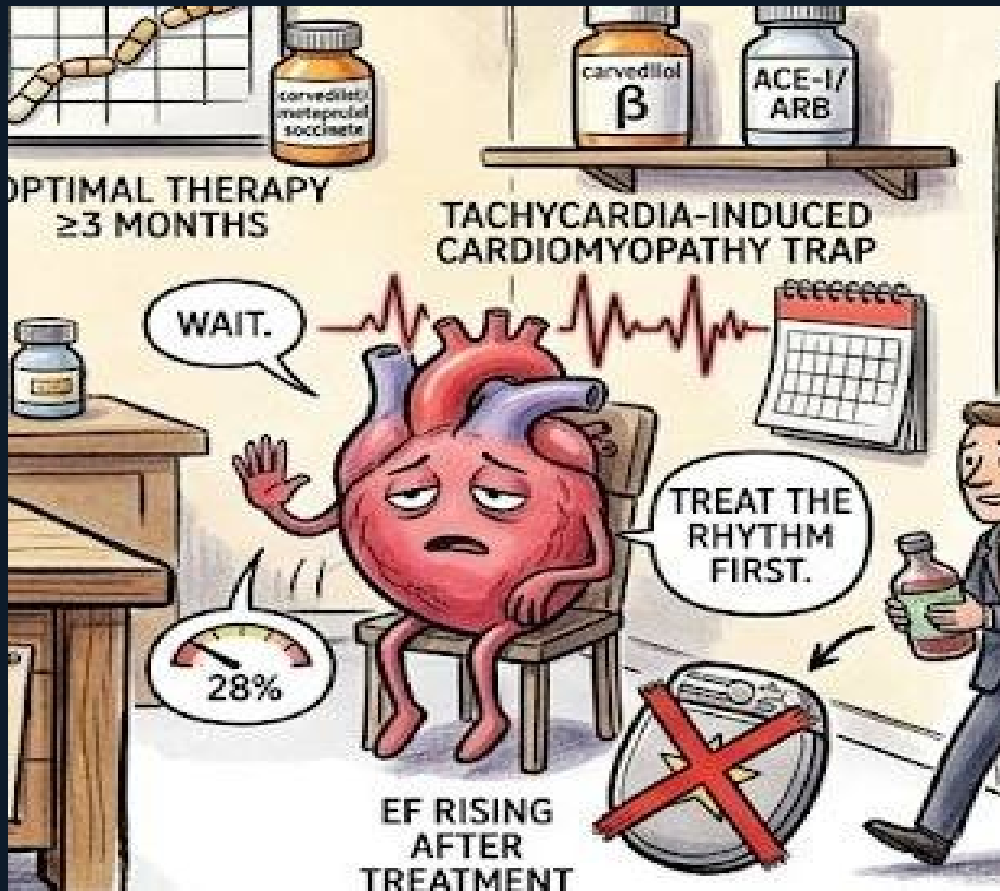


Primary Prevention — Three Gears

Three interlocking gears: EF $\leq 35\%$ (meter dial), NYHA class II–III (ladder with rungs II and III highlighted), and Optimal Therapy ≥ 3 Months (calendar with medication trail). All three must turn together.

Primary prevention ICD criteria: EF $\leq 35\%$ + NYHA II–III + on optimal medical therapy (beta-blocker + ACE-I/ARB) for ≥ 3 months. For ischemic CMP: must also be >40 days post-MI. For non-ischemic: ≥ 3 months on therapy.

Board trap: 'EF 38% $\rightarrow$ ICD indication' — wrong. Threshold is $\leq 35\%$. Trap 2: implanting ICD immediately after MI — must wait ≥ 40 days for cardiac remodeling.

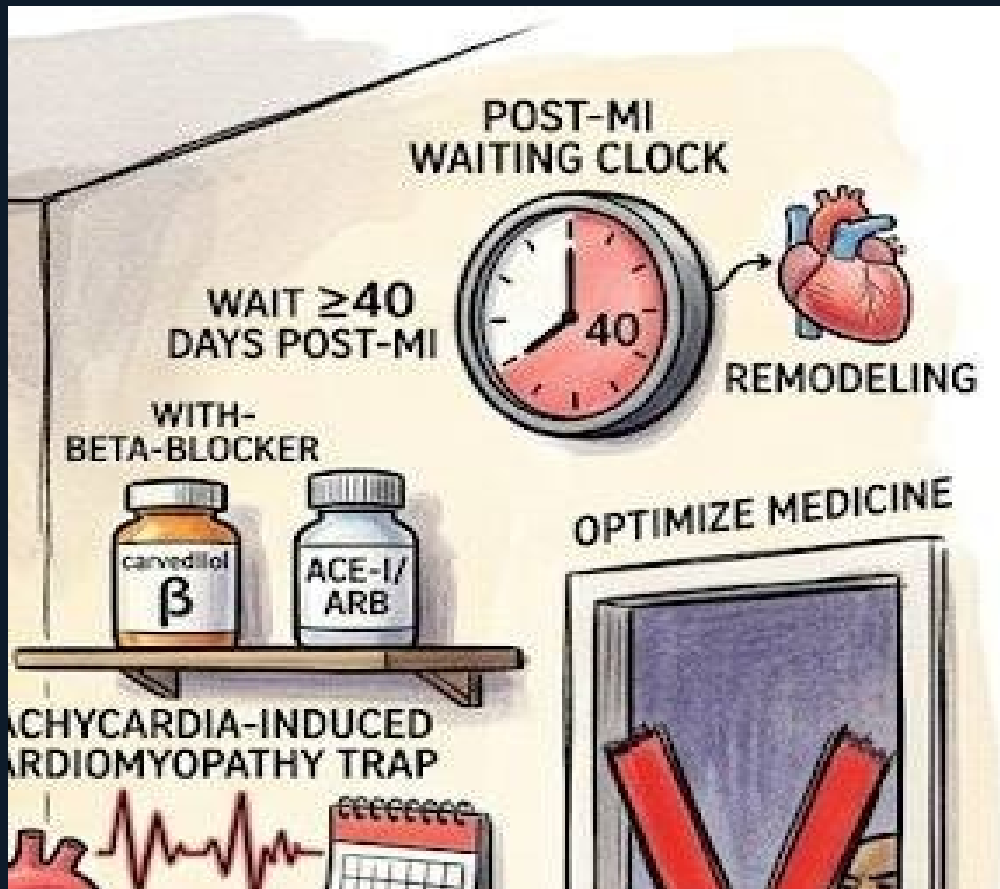


Tachycardia-Induced CMP — Wait

A dilated heart (EF 28%) is hammered by rapid tachycardia ECG waves like a pneumatic drill. The craftsman holds up his hand: 'WAIT. TREAT THE RHYTHM FIRST.' A calendar shows a waiting period then 'EF RISING AFTER TREATMENT.'

Tachycardia-induced cardiomyopathy: persistent rapid AF or incessant VT causes dilated CMP with low EF. When the tachycardia is treated (rate controlled, ablated), the EF often recovers to normal.

Board trap: 'patient with AF at 150 bpm, EF 30% → implant ICD' — wrong. Control the rate or restore rhythm first. Reassess EF after ≥3 months. Only implant ICD if EF remains ≤35% despite treatment.

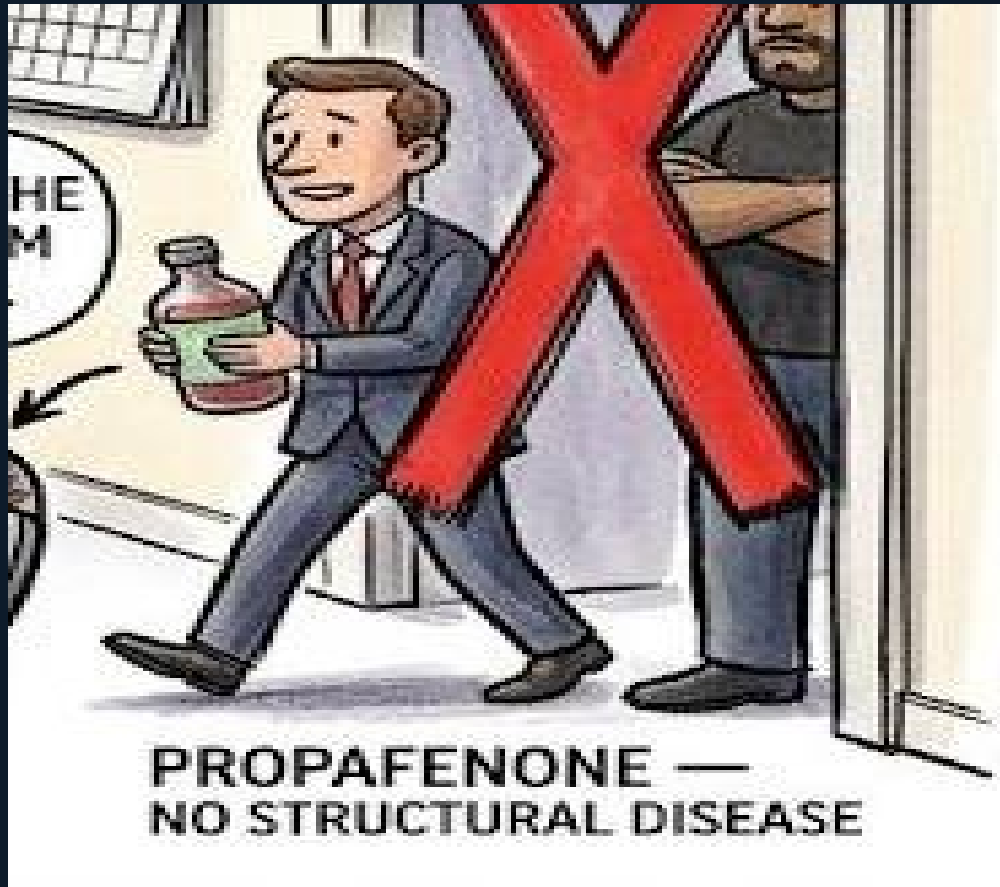


Post-MI Waiting Period

A clock shows '≥40 DAYS POST-MI' with a remodeling heart icon. During this window, a beta-blocker (carvedilol β) and ACE-I/ARB sit on the shelf doing the work.

After MI, the myocardium undergoes remodeling for up to 40 days — EF measured in this window may underestimate the final recovered EF. ICD implanted too early captures a transiently depressed EF that may recover.

Optimal medical therapy during the wait: carvedilol or metoprolol succinate + ACE-I or ARB + aldosterone antagonist if tolerated. Reassess EF at >40 days. If still $\leq 35\%$ + NYHA II–III → ICD.



Propafenone — Wrong Drug, Wrong Patient

A propafenone salesman is blocked at the workshop door by a bouncer with a red X. Label: 'PROPAFENONE — NO STRUCTURAL DISEASE.'

Class IC antiarrhythmics (propafenone, flecainide) are proarrhythmic on structurally abnormal myocardium. In patients with cardiomyopathy, prior MI, or significant HF, these drugs increase the risk of fatal ventricular arrhythmias.

Board trap: 'use propafenone for VT suppression in a patient with ischemic cardiomyopathy' — wrong. For VT in structural heart disease: amiodarone, beta-blockers, or ICD. Class IC never.

Scene E — Board Trap Master Summary

Secondary prevention: survived VF/VT → ICD

Regardless of EF; exclude reversible causes first

Primary prevention: EF ≤35% + NYHA II–III

+ Optimal therapy ≥3 months

Tachycardia-induced CMP: treat rhythm first

EF may recover; reassess before ICD

Propafenone in structural disease → fatal

Class IC only for structurally normal hearts

Reversible cause → fix first, no ICD

Acute MI <48h, electrolytes, drugs

EF 38% ≠ ICD indication

Threshold is ≤35%, not <40%

Post-MI: wait ≥40 days

Remodeling phase; EF may recover

All 3 gears must turn simultaneously

EF + NYHA + duration of optimal therapy